

R3 — ML Research Track, Week 1

Member: _____

Project setup (do first)

1. Fork the repo: <https://github.com/slvDev/esp32-ai>
2. Clone your fork. Add the original as upstream.
3. Read README.md, RESULTS.md, and research/tinystories/README.md.
4. Write 5 lines in the shared Drive doc: what you think the project does and where the hard part is.

Your task this week

Goal: the team understands how the 3 arms compare.

Steps: 1. Read research/tinystories/train.py fully. It is short. 2. Write down every hyperparameter: vocab, steps, target-core, seed, batch, sequence length, learning rate. 3. Find how the arms are sized to the same core budget. The concept is core-matching. 4. Write the 1-page explainer "How the 3 arms compare". Include a table: arm, what it tests, core size, table size.

Deliverable: the explainer as a markdown file in the club repo.

Verify: R1 and R2 read it and can explain core-matching back to you.

Learning this week

Gradient descent and loss. Watch 3Blue1Brown episode 2 again. Explain in your own words what a loss curve shows.

Resources

- train.py source: <https://github.com/slvDev/esp32-ai/blob/main/research/tinystories/train.py>
- TinyStories paper: <https://arxiv.org/abs/2305.07759>
- 3Blue1Brown neural networks: https://www.youtube.com/playlist?list=PLZHQObOWTQDNU6R1_67000Dx_ZCJB-3pi

Done when

- [] Explainer exists in the club repo
- [] R1 and R2 can explain core-matching back to you
- [] You can say what the project does in one sentence